EarningsLens

User Guide

An analyst-style read of two earnings calls,
powered by local Hugging Face models.

Version 1.0
Document type: End-user guide
Audience: Analysts, students, and reviewers

Welcome to EarningsLens

EarningsLens reads two consecutive earnings call transcripts from the same company and produces a concise, analyst-style brief. It runs five independent analyses in parallel — sentiment, hedging, topic drift, risk vocabulary, and Q&A evasion — then synthesises the structured outputs into a short narrative.

Everything runs locally on your machine using open-source Hugging Face models. Nothing is sent to a remote LLM. The output is an analytical aid: it is **not** investment advice.

What you will get

- A headline **GREEN / AMBER / RED** signal summarising the quarter-over-quarter read.
- Five diagnostic tabs you can drill into independently.
- A short, written analyst brief that grounds itself in the structured outputs.
- Provenance for every analysis so you know whether it came from the main path or a fallback.

The headline signal

GREEN	No major issues flagged across sentiment, hedging, topics, and Q&A responsiveness.
AMBER	One analytical dimension moved enough to warrant monitoring next quarter.
RED	Multiple independent stress signals appeared at once — read the transcripts closely.

Think of the signal as a triage layer. GREEN does not mean the quarter was strong — it means nothing in the language warranted special attention. RED does not mean the company is in trouble — it means several independent linguistic stress markers moved together and a human should look closely.

Getting started

1. Requirements

- Python 3.11
- Around 1 GB of disk for the first run (FinBERT ~440 MB, MiniLM ~80 MB, the local instruct model ~1 GB).
- 8 GB of RAM is enough thanks to the lightweight default model (*SmolLM2 1.7B*).
- An Alpha Vantage API key, if you want to load real tickers directly inside the app.

2. Configure your .env

Copy **.env.example** to **.env** at the project root and fill in the two variables below. The file is read at startup; restart the app after editing it.

Variable	Required?	What it does	Default / example
`MODEL`	Optional	Hugging Face model id for the local instruct LLM used by topic extraction, Q&A evasion scoring, and brief synthesis.	HuggingFaceTB / SmolLM2-1.7B-Instruct
`ALPHA_VANTAGE_API_KEY`	If using search	Free Alpha Vantage key used by **Search company** mode to fetch real ticker transcripts. Not needed for **Use sample** mode.	e.g. *ABCD1234 EFGH5678*

```
MODEL=HuggingFaceTB/SmolLM2-1.7B-Instruct
ALPHA_VANTAGE_API_KEY=your_alpha_vantage_key
```

Grab a free Alpha Vantage key at *alphavantage.co/support/#api-key*. Without it you can still use **Use sample** mode — only ticker search needs the key.

3. Install and run

```
python3.11 -m venv .venv
source .venv/bin/activate   # Windows: .venv\Scripts\activate
pip install -r requirements.txt
cp .env.example .env
python scripts/make_samples.py
streamlit run app.py
```

On macOS you can also double-click **launch.command** from Finder. The first launch downloads the three models — expect a few minutes — and subsequent launches start in seconds thanks to the on-disk model cache.

4. Optional health checks

```
./.venv/bin/pytest -q
./.venv/bin/python scripts/health_check.py
```

These verify that parsing, scoring, and model loading behave as expected before you trust any output.

Using the app

Input modes

EarningsLens supports two input modes, chosen from the sidebar:

- **Search company.** Type an exact ticker, select a match, load the available quarters, and pick the two you want to compare. Transcripts are fetched via Alpha Vantage.
- **Use sample.** Run on the built-in synthetic demo pairs — perfect for trying the tool offline or for grading.

Sidebar controls

- **Thresholds** for sentiment gap, hedging spike, and topic drift — adjust to your tolerance.
- **Q&A confidence filter** — drops low-confidence parsed Q&A pairs so the evasion tab stays clean.
- **Per-analysis timeout** — caps how long any single analyser can run before falling back.
- **Show diagnostics** — surfaces parsed speaker roles, turn segmentation, and analysis provenance.

The five analyses

Tab	What it measures	What to look at
Sentiment	FinBERT tone of prepared remarks vs. tone during Q&A.	A widening negative gap means the unscripted portion was darker than the script.
Hedging	Use of uncertainty words (may, could, pressure, headwind…) normalised per turn.	A sharp quarter-over-quarter rise often signals weaker conviction.
Topics	Semantic similarity of prepared remarks via MiniLM + themes extracted by the local LLM.	New, risk-heavy themes and dropped themes are highlighted.
Risk vocab	Curated finance vocabulary (inflation, liquidity, competition, backlog…).	Biggest movers surface first — useful for narrative shifts.
Q&A Evasion	LLM-as-judge rubric scoring whether each answer addressed its question.	Low scores flag exchanges that a human should re-read, not proof of misconduct.

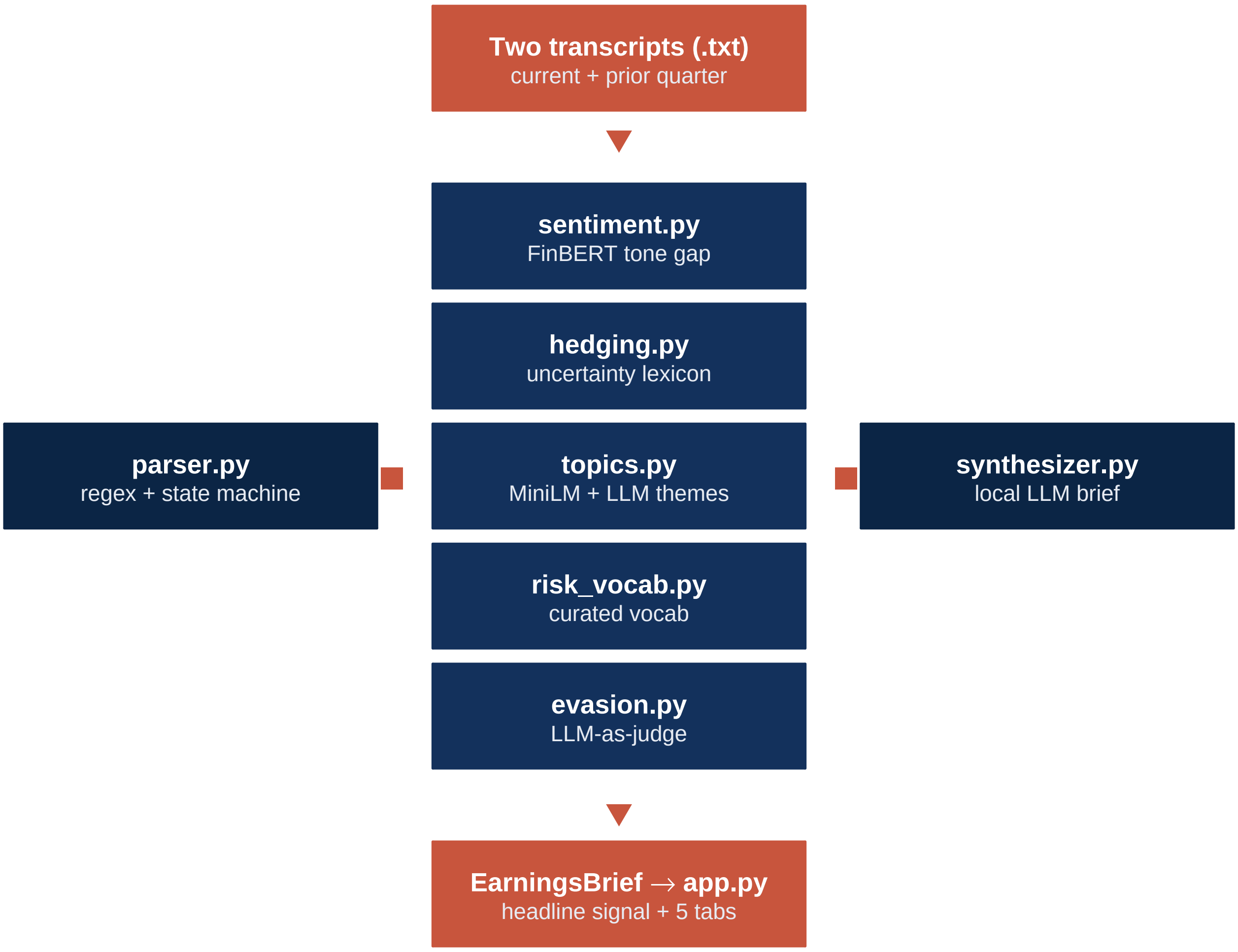
Reading the analyst brief

The brief at the top of the page is generated by the local instruct model from the structured outputs above — not from the raw transcripts. That keeps the synthesis anchored to explicit evidence and reduces the risk of hallucinated quotes. If a section fell back, the brief will say so via the provenance badge.

Under the hood

EarningsLens is intentionally layered so each signal can be inspected and trusted independently. Rule-based components are transparent; ML components are scoped narrowly.

Pipeline



Why this mix of methods

- **Parsing** is regex + a small state machine. The transcript format is repetitive enough that rules are more reliable than an LLM.
- **Sentiment** uses FinBERT because general-purpose sentiment misreads financial language (e.g. *“flat margins”*).
- **Hedging & risk vocab** are explicit lexicons. The signal is fully auditable and runs instantly on CPU.
- **Topics** mix MiniLM embeddings (broad drift) with the local LLM (human-readable themes).
- **Evasion** uses an LLM-as-judge rubric on a local model so no transcript ever leaves your machine.

Local model

The default local instruct model is **HuggingFaceTB/SmolLM2-1.7B-Instruct**. It is lightweight enough to run on an 8 GB Mac while still capable on short structured tasks: theme extraction, Q&A rubric scoring, and brief synthesis. You can swap it by editing the model id in **.env**.

Troubleshooting & tips

First run is slow

That is expected — three models are being downloaded and cached. Subsequent runs reuse the cache and are much faster.

An analysis shows a "fallback" badge

The main model path exceeded the per-analysis timeout or raised an error, so a simpler fallback path was used. The headline signal still works; treat the fallback section as lower-confidence and consider raising the timeout in the sidebar.

Q&A pairs look mismatched

Turn the Q&A confidence filter up. Low-confidence pairs are usually moderator interjections or paste artefacts in the source transcript.

Alpha Vantage rate limits

The free tier is limited. If a fetch fails, wait a minute and retry, or fall back to the sample mode while you iterate.

A note on responsible use

EarningsLens surfaces *linguistic* signals — sentiment shifts, hedging creep, topic drift, low-responsiveness Q&A exchanges. None of these prove anything on their own. They are starting points for human reading, not conclusions. The banner at the top of the app makes the same point: the output is an analytical aid, not investment advice.

EarningsLens · User Guide · v1.0 · Generated locally — no transcript data leaves your machine.